

MarketsGrid

An institutional-grade AG-Grid customization platform purpose-built for capital markets trading desks.

Profile-driven personalization, calculated columns with a real expression engine, conditional styling, enterprise persistence, and a cockpit-style settings UI — all behind a single `<MarketsGrid>` component.

AT A GLANCE

Everything a trading-desk grid needs, in one component

MarketsGrid layers a profile-driven customization platform on top of AG-Grid Enterprise. Traders save and switch full grid personalities; analysts author calculated columns, conditional styling, and reusable column templates; operators get enterprise-grade persistence, role-based config, and OpenFin-aware popouts — without bespoke wiring per desk.

Profile-driven personalization

Full grid state — columns, filters, sort, groups, pivot, viewport, sidebar — saved per profile and switched in a click. Import / export, clone, dirty tracking, and OpenFin per-view active profile resolution out of the box.

Live calculated & conditional

Real expression engine with math, stats, string, date, and logic functions. Virtual columns, row- and cell-level rules, header pulses, indicator icons, and Excel-format aware value formatters — all authored from the cockpit.

Enterprise-ready persistence

Pluggable storage adapters: in-memory, Dexie / IndexedDB, or REST-backed config service with role-based identity. Auto-save with debounce, explicit save semantics on grid state, and round-trip safe profile import / export.

9

CUSTOMIZATION
MODULES

40+

EXPRESSION FUNCTIONS

30+

FORMAT PRESETS

100%

PROFILE ROUND-TRIP

Built for: fixed-income, equity, FX, and credit desks running OpenFin, Workspace, or plain-browser deployments. Drop-in for AG-Grid Enterprise customers; pairs with [@starui/config-service](#) for multi-app, multi-tenant config governance.

Feature inventory

The pages that follow are a code-derived inventory of every public surface, module, and capability shipped by MarketsGrid in the current build — compiled by reading the source under `packages/markets-grid/`, `packages/core/src/modules/`, `packages/core/src/expression/`, `packages/core/src/ui/`, and `packages/core/src/profiles/`. No marketing prose — just what the code actually does.

1. `<MarketsGrid>` — public component

Props (`MarketsGridProps`)

- `gridId` — keys profile storage per instance
- `rowData` — row array
- `columnDefs` — base column definitions
- `modules` — module list (defaults to `DEFAULT_MODULES`)
- `theme` — AG-Grid theme object
- `rowIdField` — `string | string[]` , defaults to `'id'`
- `rowHeight` , `headerHeight`
- `animateRows`
- `defaultColDef`
- `sideBar` — `SideBarDef | boolean`
- `statusBar`
- `showToolBar` , `showFiltersToolBar` , `showFormattingToolBar`
- `showSaveButton` , `showSettingsButton` , `showProfileSelector`
- `storageAdapter` — single `StorageAdapter` instance
- `storage` — `StorageAdapterFactory` (closes over `appId / userId` ; precedence over `storageAdapter`)
- `instanceId` — stable per-instance identity (defaults to `gridId`)
- `appId` , `userId`
- `autoSaveDebounceMs` (default 300)
- `onGridReady` — passthrough
- `onReady` — fires once after AG-Grid ready + platform mount + active profile applied
- `adminActions` — entries in Settings Tools menu
- `gridLevelData` + `onGridLevelDataLoad` — opaque per-grid blob
- `headerExtras` — `ReactNode` slot above toolbars
- `className` , `style`

`MarketsGridHandle` (imperative ref)

- `gridApi` — AG-Grid `GridApi`
- `platform` — `GridPlatform`
- `profiles` — `UseProfileManagerResult`

Public types exported from `packages/markets-grid/src`

- `SavedFilter` — { `id`, `label`, `filterModel`, `active` }
- `AdminAction` — { `id`, `label`, `icon?`, `description?`, `onClick`, `visible?` }
- `StorageAdapterFactory`, `StorageAdapterFactoryOpts`

2. Host-chrome subcomponents (composed by `<MarketsGrid>`)

- `FiltersToolbar`
 - Saved-filter pill carousel
 - Per-pill row-count badge
 - Add / toggle / rename / clear-all / + (delta-aware)
 - Carousel collapse → summary chip; persisted via `toolbar-visibility`
 - Sticky add / clear-all action cluster
- `FormattingToolbar`
 - Inline-pinned + draggable / pop-out variants
 - Modules: Context · Type · Format · Paint · Clear · Library
 - Live sample, undo/redo, rule-trail preview
 - `style-toolbar-toggle` Brush pill on filters row
- `SettingsSheet`
 - Cockpit drawer; 3-col layout (module rail / items list / editor)
 - Renders flat `SettingsPanel` or master-detail `ListPane` + `EditorPane`
 - `<PopoutPortal>` support (separate OS window)
 - Per-module back-compat `data-testid`s
- `ProfileSelector`
 - Dropdown + create / load / delete / clone / rename
 - Per-row Export, footer Export-active + Import
 - Dirty indicator on active profile
- Save button — captures grid state via `captureGridStateInto`, calls `profiles.save()`, dirty-dot, success flash
- Settings button — opens sheet inline or refocuses popped window

3. Modules in `DEFAULT_MODULES`

Wired in priority order; each ships in every profile snapshot.

#	MODULE ID	PRIORITY	SCHEMA	HAS SETTINGSPANEL
1	general-settings	0	v3	yes
2	column-templates	5	v1	no
3	column-customization	10	v5	yes (master-detail)
4	calculated-columns	15	v1	yes (master-detail)
5	column-groups	18	v1	yes (master-detail)
6	conditional-styling	20	v1	yes (master-detail)
7	grid-state	200	v2	no
8	toolbar-visibility	1000	v1	no
9	saved-filters	1001	v1	no

4. general-settings — Grid Options panel

Essentials

- `rowHeight`, `headerHeight`, `animateRows`
- `quickFilterText`
- Pagination — `enabled`, `pageSize`, `autoPageSize`, `hide-panel`
- Selection — `mode`, `checkbox`, `cellSelection`, `row dragging`
- Cell flash + fade duration

Row grouping / pivot

- `groupDisplayType`, `groupDefaultExpanded`
- `rowGroupPanelShow`, `rowGroupPanelSuppressSort`
- `groupHideOpenParents`, `groupHideColumnsUntilExpanded`
- `groupShowOpenedGroup`, `groupRemoveSingleChildren` | `leafGroupsOnly`
- `groupAllowUnbalanced`, `groupMaintainOrder`
- `stickyGroups`, `groupLockGroupColumns`
- `groupDragLeaveHidesColumns`
- `suppressGroupChangesColumnVisibility` (4-way)
- `refreshAfterGroupEdit`, `serverSideExpandAllAffectsAllRows`
- `pivotMode`, `pivotPanelShow`
- `grandTotalRow`, `groupTotalRow`, `suppressAggFuncInHeader`

Filter / sort / clipboard

- `enableAdvancedFilter`, `includeHiddenColumnsInQuickFilter`
- Multi-sort mode (compound → `suppressMultiSort` + `alwaysMultiSort` + `multiSortKey`)
- `accentedSort`
- `copyHeadersToClipboard`, `clipboardDelimiter`

Editing / interaction

- `singleClickEdit`, `stopEditingWhenCellsLoseFocus`
- Enter navigation (compound → vertical + after-edit)
- `undoRedoCellEditing` + limit
- `tooltipShowDelay`, `tooltipShowMode`

Styling

- `suppressRowHoverHighlight`, `columnHoverHighlight`
- `suppressCellTextSelection`, `suppressColumnMoveAnimation`

Default colDef (7 subsections)

- SIZING — `resizable`, `minWidth` / `maxWidth` / `width` / `flex`, `suppressSizeToFit`, `suppressAutoSize`
- SORT & FILTER — `sortable`, `filter`, `unSortIcon`, `floatingFilter`
- EDITING — `editable`, `suppressPaste`, `suppressNavigable`
- HEADER — `wrapHeaderText`, `autoHeaderHeight`, `suppressHeaderMenuButton`
- MOVEMENT & LOCKING — `suppressMovable`, `lockPosition`, `lockVisible`, `lockPinned`
- CELL CONTENT — `wrapText`, `autoHeight`, `enableCellChangeFlash`
- GROUPING · PIVOT · VALUES — `enableRowGroup`, `enablePivot`, `enableValue`

Sidebar

- Visibility, per-panel toggles (columns / filters / custom), default panel

Status bar

- Visibility + per-panel options

5. `column-templates` — reusable override bundles

- `templates`: `Record<id, ColumnTemplate>`
- `typeDefaults`: `Partial<Record<ColumnDataType, templateId>>` (`numeric` / `date` / `string` / `boolean`)
- Template fields — `id`, `name`, `description?`, `cellStyleOverrides`, `headerStyleOverrides`, `valueFormatterTemplate`, `sortable`, `filterable`, `resizable`, `cellEditorName`, `cellEditorParams`, `cellRendererName`, `createdAt`, `updatedAt`
- Resolution — per-field merge for style overrides; last-writer-wins elsewhere; opaque `cellEditorParams`; `typeDefaults[dataType]` folds at chain bottom unless `templateIds`: `[]`
- Authored from Formatting Toolbar (save) + Column Settings TEMPLATES band (apply / remove)
- No `SettingsPanel`; no `transformColumnDefs` — passive state holder

6. `column-customization` — per-column overrides

State

- `assignments`: `Record<colId, ColumnAssignment>`

ColumnAssignment fields

- `headerName`, `headerTooltip`
- `initialWidth`, `initialHide`, `initialPinned`
- `cellStyleOverrides`, `headerStyleOverrides`
- `valueFormatterTemplate`
- `filter`: `ColumnFilterConfig`
- `rowGrouping`: `RowGroupingConfig`
- `editable`
- `templateIds`: `string[]`

ColumnFilterConfig

- Tri-state enable
- Kind picker — `agTextColumnFilter` / `agNumberColumnFilter` / `agDateColumnFilter` / `agSetColumnFilter` / `agMultiColumnFilter`
- Floating-filter switch
- Buttons multi-select — `apply` / `clear` / `reset` / `cancel`
- `debounce`, `closeOnApply`
- Set-filter sub-controls — mini-filter, select-all, alphabetical sort, Excel mode (Windows / Mac), default-to-nothing
- Multi-filter sub-list — ordered sub-filters with display-mode (`inline` / `subMenu` / `accordion`)

RowGroupingConfig

- `enableRowGroup`, `enableValue`, `enablePivot` (tool-panel)
- `rowGroup` + index, `pivot` + index (initial)
- Agg function — `sum` / `min` / `max` / `count` / `avg` / `first` / `last` / `custom` (expression evaluated via shared engine, `[value]` exposed)
- Grid-level: `groupDisplayType`, `groupTotalRow`, `grandTotalRow`, `suppressAggFuncInHeader`

Reducers (`packages/core/src/modules/column-customization/`)

- `applyTypographyReducer` — `fontFamily`, `fontSize`, `fontWeight`, `italic`, `alignment`
- `applyColorsReducer` — `foreground`, `background`
- `applyAlignmentReducer` — `horizontal`, `vertical`
- `applyBordersReducer`, `clearAllBordersReducer`
- `applyHeaderNameReducer`, `applyEditableReducer`
- `applyFormatterReducer`
- `applyTemplateToColumnsReducer`, `removeTemplateRefFromAssignmentsReducer`
- `clearAllStylesReducer`, `clearAllStylesInProfileReducer`
- `writeOverridesReducer` (cell vs header)
- `mergeOverrides`, `stripUndefined`, `overrideKey`

Pipeline

- Header alignment falls back through `headerStyleOverrides` → `cellStyleOverrides`
- CSS rule injection via `ResourceScope` (cleaned up on unmount)
- Per-column classes `.gc-col-c-{colId}` / `.gc-hdr-c-{colId}`

Panel

- ListPane with windowing past 60 columns (36px row, 8-row overscan)
- Override-badge memoized as `Set<colId>`; single `useDirtyColIds()` subscription

7. `calculated-columns` — virtual columns

State

- `virtualColumns: VirtualColumnDef[]` (`id`, `headerName`, `expression`, `dataType?`, `valueFormatterTemplate?`)

Behaviour

- Virtual columns appended via `transformColumnDefs`; appear in `api.getColumns()`
- Cross-row `EvaluationContext.allRows` populated from per-`GridApi` `WeakMap`
- Cache invalidates on `rowDataUpdated` / `modelUpdated` / `cellValueChanged`
- Aggregate refresh via `api.refreshCells({ columns: virtualColIds, force: true })` on edits
- Inherit column-customization styling pipeline + Excel `[Red]` / `[Green]` colour resolver
- Header alignment fallback chain applies to virtuals
- Picked up by Column Groups composer
- Compatible with `column-customization` filter / grouping configs

Panel

- Master-detail with Monaco `ExpressionEditor` (column / function autocomplete, `[col]` hints)
- Shared `FormatterPicker` for value formatter

8. `column-groups` — nestable group editor

State

- `groups: ColumnGroupNode[]` — children are `{ kind: 'col', colId, show? }` or `{ kind: 'group', group }`
- `openGroupIds: Record<groupId, boolean>`

Per-group fields

- `groupId` (stable), `headerName`
- `openByDefault`, `marryChildren`
- `headerStyle` — bold, italic, underline, fontSize, color, background, align, borders

Behaviour

- Runtime `columnGroupOpened` writes to `openGroupIds`; restored on transform
- Stale `openGroupIds` pruned on deserialize
- CSS injection — `gc-hdr-grp-{groupId}` class + `::after` border overlay
- Pure tree ops module — `flattenGroups`, `updateGroupAtPath`, `deleteGroupAtPath`, `moveGroupAtPath`, `findGroupByPath`
- Column visibility tri-state per chip — `always` / `open` / `closed` (maps to `columnGroupShow`)
- Add subgroup capped at depth 3 (UI rule)

9. conditional-styling — rule editor

Per-rule fields

- `id`, `enabled`, `priority`
- `expression` (Monaco)
- `scope` — `cell` (column list) or `row`
- `targetColumns: colId[]`
- `valueFormatterTemplate?` (`preset` / `excelFormat` / `expression` / `tick`)
- `flash` — `enabled`, `target` (`cells` / `headers` / `cells+headers`), `flashDuration`, `fadeDuration`, `continuous-pulse toggle`
- `indicator` — `icon` (Lucide, looked up via `INDICATOR_ICONS` / `findIndicatorIcon`), `color`, `position` (`top-left` / `top-right`), `target` (`cells` / `headers` / `both`)

Pipeline

- Cell rules emit `cellClassRules` (`gc-rule-{id}` class)
- Row rules emit `rowClassRules`
- Header pulse + indicators driven by DOM watcher (subscribes to `modelUpdated`, `filterChanged`, `cellValueChanged`, rule-state change)
- Indicators painted via CSS `::before`, excluding `.ag-floating-filter`

10. grid-state — native AG-Grid state persistence

Captured (explicit Save only)

- Full `api.getState()` — column order / visibility / width / pinning / sort / filter model / column-group open-closed / pagination / sidebar / focus / selection / row-group expansion
- Viewport anchor — `firstRowIndex`, `leftColId`, `horizontalPixel`
- `quickFilter`

Replay

- On `onGridReady` (cold mount) and `profile:loaded`
- Empty / new profile clears via `api.setState({})` + clear `quickFilterText`
- Selection column re-pinned + reordered post- `setState` via `applyColumnState` deferred to microtask + `firstDataRendered`
- Stale saved order merged with live column set (saved IDs first, then new IDs appended)

11. saved-filters

- Opaque host-defined state (`filters: unknown[]`)
- Host shape (`markets-grid/src/types.ts`): `{ id, label, filterModel, active }`
- No transforms / `SettingsPanel`; ships in profile snapshot for round-trip
- Pure logic in `filtersToolbarLogic.ts` — `generateLabel`, `doesRowMatchFilterModel`, `filterModelsEqual`, `mergeFilterModels`, `subtractFilterModel`, `isNewFilter`

12. toolbar-visibility

- `visible: Record<toolbarId, boolean>` (missing key = host default)
- Drops non-boolean values on deserialize
- First wired consumer — FiltersToolbar collapse state under key `filters-toolbar-pills`

13. Expression Engine — built-in functions

Math

- `ABS`, `ROUND`, `FLOOR`, `CEIL`
- `SQRT`, `POW`, `MOD`, `LOG`, `EXP`

Aggregation (column-ref via `[col]`)

- `SUM`, `COUNT`, `DISTINCT_COUNT`
- `MIN`, `MAX`

Statistical (column-ref via `[col]`)

- `AVG`, `MEDIAN`, `STDEV`, `VARIANCE`

String

- `CONCAT`, `UPPER`, `LOWER`, `TRIM`
- `SUBSTRING`, `REPLACE`, `LEN`
- `STARTS_WITH`, `ENDS_WITH`, `CONTAINS`, `REGEX_MATCH`

Date

- `NOW`, `TODAY`
- `YEAR`, `MONTH`, `DAY`, `IS_WEEKDAY`
- `DATE_DIFF` (units: days/d, hours/h, minutes/m, seconds/s)
- `DATE_ADD` (units: days/d, months/mo, years/y, hours/h)

Logic

- `IF`, `IFS`, `SWITCH`, `CASE` (alias of `SWITCH`)
- `ISNULL`, `ISNOTNULL`, `ISEMPTY`

Engine exports

- `ExpressionEngine`, `tokenize`, `parse`, `Evaluator`
- `tryCompileToAgString`
- `migrateExpressionSyntax`, `migrateExpressionsInObject`

14. FormatterPicker — preset catalogue

Number (`num-*`)

- `num-integer` , `num-2dp` , `num-4dp`
- `num-neg-parens` , `num-neg-red-parens`
- `num-green-red-nosign` , `num-green-red-usd`
- `num-scientific` , `num-bps`

Tick (`fixed-income`, `tick-*`)

- `tick-32` , `tick-32-plus`
- `tick-64` , `tick-128` , `tick-256`
- `TickToken` constants — `TICK32` , `TICK32_PLUS` , `TICK64` , `TICK128` , `TICK256`

Currency (`cur-*`)

- USD: `cur-usd` , `cur-usd-red-neg` , `cur-usd-parens` , `cur-usd-green-red-nosign`
- EUR: `cur-eur` , `cur-eur-green-red-nosign`
- GBP: `cur-gbp` , `cur-gbp-green-red-nosign` (SSF-quoted)
- JPY: `cur-jpy` , `cur-jpy-green-red-nosign` (no decimals)
- INR: `cur-inr` , `cur-inr-green-red-nosign` (SSF-quoted)

Percent (`pct-*`)

- `pct-0` , `pct-2` , `pct-bps`

Date (`date-*`)

- `date-iso` , `date-us` , `date-eu` , `date-long`

DateTime (`dt-*`)

- `dt-iso` , `dt-us-short` (+ all date presets re-exposed)

String (`str-*`)

- `str-default` , `str-suffix-units`

Boolean (`bool-*`)

- `bool-yn` , `bool-yes-no` , `bool-check`

Helpers

- `inferPickerDataType()` , `presetsForDataType(dataType)`

Template kinds (`ValueFormatterTemplate`)

- { `kind`: 'preset', `presetId` }
- { `kind`: 'excelFormat', `format` }
- { `kind`: 'expression', `expression` } (gated by CSP policy)
- { `kind`: 'tick', `tick`: `TickToken` }

15. Cockpit / SettingsPanel UI primitives

packages/core/src/ui/SettingsPanel/ :

- PanelChrome , Band , FigmaPanelSection
- SubLabel , ObjectTitleRow , TitleInput
- ItemCard , PairRow
- IconInput , Stepper
- PillToggleGroup , PillToggleBtn
- SharpBtn (default / action / ghost / danger)
- TGroup , TBtn , TDivider
- MetaCell , GhostIcon
- DirtyDot , LedBar
- Caps , Mono
- TabStrip
- CockpitList , CockpitListItem (cmdk-backed)

16. Format-editor primitives

packages/core/src/ui/format-editor/ :

- FormatPopover , FormatColorPicker , FormatDropdown
- BorderSidesEditor (with BorderSide , BorderStyle , BorderMode , SideSpec types)
- ExcelReferencePopover
- registerPopoverRoot , clickIsInsideAnyOpenPopover (popover stack)

17. StyleEditor sections

- TextSection , ColorSection , BorderSection , FormatSection
- BorderStyleEditor
- Opt-in sections={ [...]} API

18. ProfileManager — public methods

packages/core/src/profiles/ProfileManager.ts :

Lifecycle

- boot() — resolve active id (ActiveIdSource → localStorage → reserved Default), load snapshot, wire auto-save / dirty tracker
- dispose()

State

- subscribe(listener) → unsubscribe
- getState() → { activeId , profiles[] , isLoading , isDirty }

Operations

- `save()` — flush debounce + persist
- `discard()` — reload from disk + clear dirty
- `load(id, { skipFlush? })`
- `create(name, { id? })`
- `remove(id)` (Default cannot be deleted)
- `rename(id, name)`
- `clone(sourceId, name, { id? })`

Import / Export

- `export(id?)` → `ExportedProfilePayload`
- `import(payload, { name?, activate?, sanitize? })`

Identity sources

- Pluggable `ActiveIdSource` (e.g. `createOpenFinViewProfileSource()` for per-view active profile in OpenFin)

Constants

- `RESERVED_DEFAULT_PROFILE_ID`, `activeProfileKey`

19. Storage adapters

- `StorageAdapter` interface — `loadProfile`, `saveProfile`, `deleteProfile`, `listProfiles`, `loadGridLevelData?`, `saveGridLevelData?`
- `MemoryAdapter` — ephemeral
- `DexieAdapter` — IndexedDB
- `StorageAdapterFactory` — `(opts: { instanceId, appId?, userId? }) => StorageAdapter`
- `ConfigService` factory (separate package `@starui/config-service`) — bundles all profiles into one `AppConfigRow` with `componentType: 'markets-grid-profile-set'`

20. AdminAction (Settings Tools menu)

- `id`, `label`, `onClick`
- `icon?` (Lucide refs: `database`, `file-text`, `list-checks`, `activity`, `bar-chart-3`, `shield-check`, `users`, `terminal`, `eye`, `wrench`, `refresh-cw`; defaults to `wrench`)
- `description?`, `visible?`
- Test id `admin-action-${id}`

21. Expression security policy

```
configureExpressionPolicy({ mode, onViolation }):
```

- Modes — `allow` (default) / `warn` / `strict`
- Strict mode — `valueFormatterFromTemplate` returns identity for `kind: 'expression'`; `ProfileManager.import` rejects expression-kind templates
- `import({ sanitize: true })` rewrites expression templates to `{ kind: 'preset', presetId: 'num-integer' }` (or equivalent)
- Two enforcement points — runtime compile + import-time scan
- `onViolation({ kind, expression, reason })` observer; errors swallowed
- `getExpressionPolicy()` reader

22. Core platform exports (`@starui/core`)

Platform

- `GridPlatform`, `EventBus`, `ApiHub`, `ResourceScope`, `CssInjector`, `PipelineRunner`, `topoSortModules`

Hooks

- `GridProvider`, `useGridPlatform`
- `useModuleState`, `useModuleDraft`
- `useGridApi`, `useGridEvent`, `useGridColumnns`
- `useProfileManager`
- `useDirty`, `useDirtyCount`
- `useUndoRedo`, `HistoryStack`

Popout / portals

- `PopoutPortal`, `Poppable`
- `PortalContainerProvider`, `usePortalContainer`
- `openFinWindowOpener`, `isOpenFin`

Editors

- `ExpressionEditor` (Monaco)

CSS

- `cockpitCSS`, `COCKPIT_STYLE_ID`
- `v2SheetCSS`, `V2_SHEET_STYLE_ID` (aliases)

shadcn re-exports

- `Button`, `Input`, `Textarea`, `Select`, `Switch`
- `Popover`, `AlertDialog`, `Tooltip`
- `Separator`, `Label`, `ToggleGroup`, `ColorPicker`

23. Cross-cutting invariants (from code)

- Single source of truth — IndexedDB profile snapshots (`gc-customizer-v2`)
 - Per-module `schemaVersion` with optional `migrate(raw, fromVersion)`
 - Save path — Save click → `captureGridStateInto` → `serializeAll` → `persist` → `profile:saved` event
 - Auto-save 300ms debounce, opt-in (`disableAutoSave: true` is the production default)
 - `grid-state` only writes on explicit Save
 - Cross-module state read exclusively via `ctx.getModuleState<T>(moduleId)` — modules never import each other
 - Module ordering: `0 → 5 → 10 → 15 → 18 → 20 → 200 → 1000 → 1001`
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About this document. Compiled directly from the MarketsGrid source tree. Sections are kept in lockstep with `MarketsGrid.DEFAULT_MODULES`, the cockpit settings primitives, and the public exports of `@starui/core`. For the canonical, machine-readable source see `docs/2026-05-07/architecture-and-design/MARKETS_GRID_FEATURES.md` in the MarketsUI platform repository.